# Testing Time Dilation: Muon Lifetimes and the Demise of the Ether

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## Introduction

When Einstein proposed special relativity in 1905, the idea that time itself is relative to an observer's motion was revolutionary and controversial. In Newton's classical framework, time flows identically for all observers. Clocks cannot disagree on durations, no matter their motion relative to each other. Lorentz, in the process of deriving the transformations that now bear his name, did not let go of the concept of an ether: a medium for light propagation from which all motion could be measured. According to Lorentz, the apparent "slowing down" of moving clocks arose because of their interaction with this hidden ether (Lorentz, 1904).

In contrast, Einstein (1905) argued that time dilation is not an illusion or a mechanical effect of a medium, but instead a geometric feature of spacetime itself. Thus the conflict is threefold. Newtonian physics asserts that time is absolute, Lorentz's theory keeps time dilation but explains it through an ether, and Einstein's relativity treats time dilation as an intrinsic spacetime property.

The experiment of Bailey et al. (1977) at CERN provides experimental data which will arbitrate between these views. Experimental data will be the basis for our determination of which model best describes the physical workings of the universe. By measuring the lifetimes of relativistic muons in a storage ring, they confirmed Einstein's predictions with high precision. Crucially, they demonstrated that acceleration does not alter time dilation. This fact undermines Lorentz's ether hypothesis.

## Newtonian Time

Newton's *Principia* defined time as "absolute, true, and mathematical, flowing equably without relation to anything external." (Newton, 1687/1999, p. 408) According to this view, decay processes that happen at a regular rate should not depend on the speed of a particle relative to an observer. A muon, which decays at $\tau_0 \approx 2.2\mu s$ in its rest frame, will decay at the same rate in all other frames, no matter how fast the muon is moving.

If Newton's picture were correct, muons created by cosmic rays in the upper atmosphere could not reach the Earth's surface. Their rest lifetime is simply too short. Even at their relativistic speeds, muons would still vanish after a few hundred meters of travel. Experiments beginning in the 1940s showed unprecedented amounts of muons reaching surface detectors, suggesting that relativistic effects of time dilation and length contraction are in effect (Rossi & Hall, 1941). Bailey et al.'s storage-ring measurements drove the point home with precision.

## Lorentz's Ether

Lorentz introduced transformations that preserve the constancy of the speed of light but insisted that these effects were the result of motion through an invisible ether. In this interpretation, time dilation is real, but it reflects how clocks interact with the ether rather than a fundamental property of spacetime.

In a circular accelerator, muons are not in pure inertial motion; they experience continuous centripetal acceleration. If Lorentz's ether was the true cause of time dilation, the acceleration could expose deviations from the simple inertial formula $\tau_{\text{lab}} = \gamma\tau_0$. Therefore, measuring whether accelerated muons live exactly as long as relativity predicts is a direct test of the ether hypothesis.

## Einstein and Minkowski's Framework

Special relativity provides a different foundation. In Minkowski spacetime, the line element

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2. \tag{1}$$

gives the invariant proper time along a particle's worldline. The proper time increment is

$$d\tau = \frac{-ds}{c^2} \tag{2}$$

$$d\tau = dt\sqrt{1 - \frac{v^2}{c^2}} = \frac{dt}{\gamma}, \tag{3}$$

where

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}. \tag{4}$$

For an unstable particle such as a muon, the rest-frame decay law is exponential:

$$N(\tau) = N_0 e^{-\tau/\tau_0}, \tag{5}$$

where $\tau_0$ is the mean proper lifetime and $\tau$ is the proper time along the muon's worldline. In the laboratory frame, this becomes:

$$N(t) = N_0 e^{-t/(\gamma\tau_0)}. \tag{6}$$

So the mean lifetime is dilated:

$$\tau_{\text{lab}} = \gamma\, \tau_0. \tag{7}$$

This prediction is geometric, not dependent on ether or absolute time, and should hold even if the particle is undergoing acceleration. Bailey et al. stored muons with momentum of about p = 3.094 GeV/$c$ corresponding to a Lorentz factor of $\gamma = 29.33$. The rest lifetime of muons is $\tau_0 \approx 2.197\mu s$. This had been measured previously through stopping muon experiments. Then relativity predicts

$$\tau_{\text{lab}} = (29.33)(2.197\ \mu s) \approx 64.45\ \mu s. \tag{8}$$

Their measurements yielded values of:

$$\tau^{\text{lab}}_{\mu^+} = 64.419(58)\ \mu s, \tag{9}$$

$$\tau^{\text{lab}}_{\mu^-} = 64.368(29)\ \mu s. \tag{10}$$

in striking agreement with theory. By dividing these values by $\gamma$, they inferred the proper muon lifetime, $\tau_{0\mu^-} = 2.1948(10)\ \mu s$.

# Implications for Skeptical Viewpoints

The results directly contradict the Newtonian model: muons' lifetime in the lab frame does differ from the muon proper decay time. Muons in the lab frame survive $\sim$30 times longer, exactly as relativity predicts. More significantly, the results also undermine Lorentz's ether view. If the ether determined time dilation, accelerated motion might have revealed deviations from the simple relativistic formula. But Bailey et al. observed no such effect: accelerated muons obey the same $\tau_{(lab)} = \gamma\tau_0$ law as inertial ones. The absence of ether caused deviations at a precision of $2 \times 10^{-3}$, strongly suggests that the ether is unnecessary and likely incorrect. Therefore, time dilation is best understood as a property of spacetime geometry, not as a consequence of a hidden medium.

## References


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